

DealPulse Scout

LangGraph multi-agent car deal finder powered by Bright
Data + Claude

Web Data Unlocked Hackathon • Pipeline v2

The Problem

- Car listings are scattered across CarGurus, AutoTrader, Craigslist, and dealer sites
- No single place answers: *"What's the best deal near me for this car under my budget?"*
- Manual comparison is slow — price, mileage, market value, and location all matter
- Buyers need ranked, explained recommendations — not just a list of links

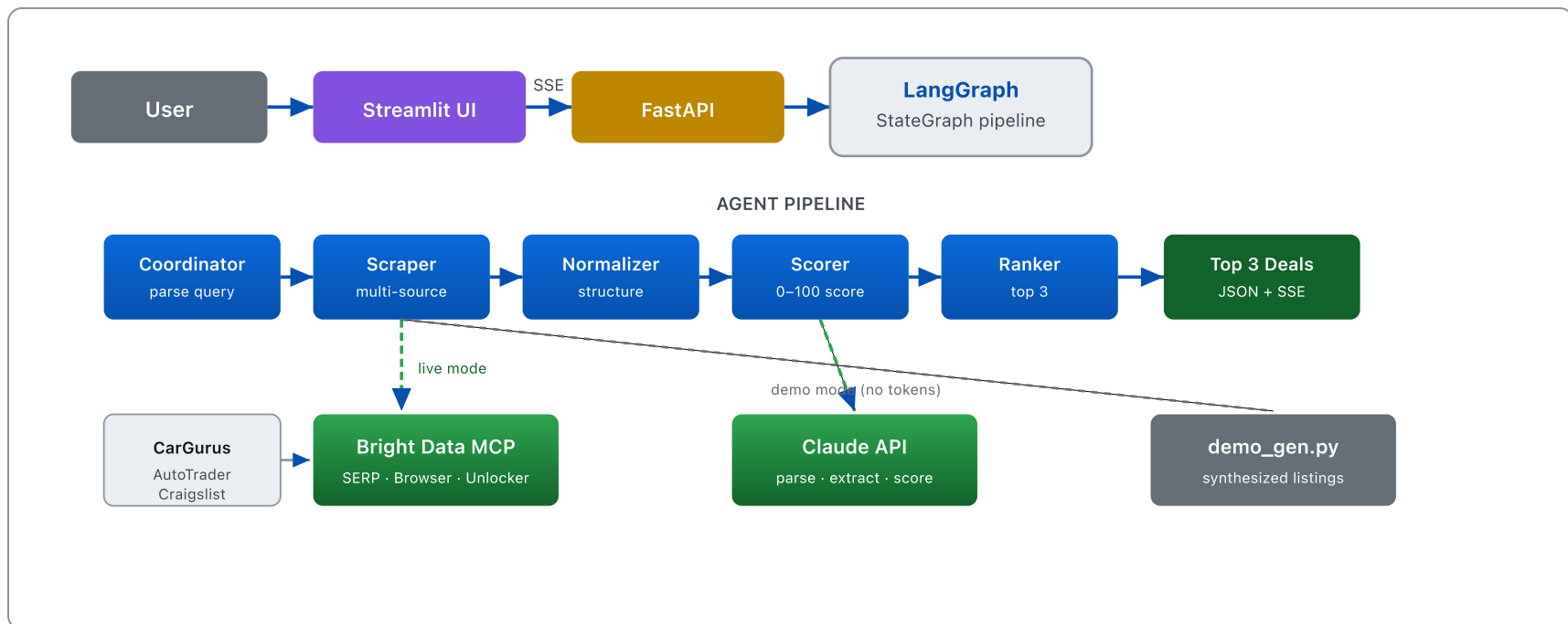
Our Solution

Type a natural-language query. Get the **top 3 ranked deals** with scores and reasons.

BMW X5 Austin under 75k
Lamborghini Huracán Miami under 300k
Porsche Panamera Austin under 120k

Five LangGraph agents coordinate scraping, normalization, scoring, and ranking — streamed live to the UI.

Architecture



Demo mode synthesizes listings; live mode scrapes via Bright Data and scores with Claude.

Bright Data Integration

SERP (search_engine)

Market price research — comparable listings and price trends for scoring context.

Scraping Browser

JS-heavy sites: CarGurus, AutoTrader — full rendered page content.

Web Unlocker

Dealer sites and Craigslist — bypass blocks, return markdown.

MCP Protocol

Agents call tools via Bright Data MCP SSE endpoint — no custom scraper infra.

Claude AI Layer

- **Coordinator** — interprets fuzzy queries ("75k", "near Austin", multi-word makes)
- **Normalizer** — extracts structured fields from raw HTML/markdown scrapes
- **Scorer** — compares listing price vs SERP market data; writes one-line reason

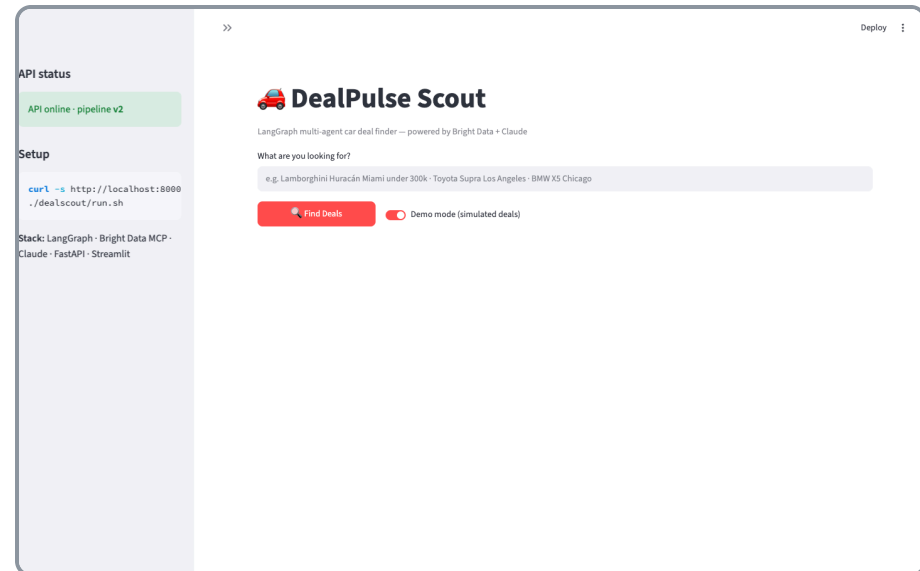
Heuristic fallbacks in demo/offline mode — pipeline never hard-crashes on stage.

Live Demo UI

Streamlit Frontend

- Natural-language search box
- Real-time agent log (SSE)
- Top 3 expandable deal cards
- One-click marketplace links

Sidebar confirms **pipeline v2** and API health.



Demo Walkthrough

Query: **BMW X5 Austin under 75K**

Rank	Deal	Score	Source
#1	2026 BMW X5	88	CarGurus
#2	2025 BMW X5 Premium	82	AutoTrader
#3	2024 BMW X5 Sport	79	Craigslist

CarGurus link uses BMW X5 entity **m3/d393** — live inventory, not a fake VIN.

API status

API online · pipeline v2

Setup

```
curl -s http://localhost:8000
./dealscout/run.sh
```

Stack: LangGraph · Bright Data MCP · Claude · FastAPI · Streamlit

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What are you looking for?

BMW X5 Austin under 75K

Find Deals

Demo mode (simulated deals)

Coordinator parsing: 'BMW X5 Austin under 75K'

Plan: BMW X5 near Austin, TX, max \$75,000

Scraper starting multi-source fetch...

DEMO_MODE — generating listings from search plan

Prepared 3 demo listings for BMW X5

Normalizer extracting structured listings...

Generated 3 demo listings for BMW X5

Scorer evaluating 3 listings...

Using pre-scored cache (3 deals)

Ranker selecting top 3 deals...

#1 2026 BMW X5 — \$58,073 (score 88.0)

#2 2025 BMW X5 Premium — \$55,169 (score 82.0)

#3 2024 BMW X5 Sport — \$52,265 (score 79.0)

Returning top 3 deals

Top 3 Deals

#1 — 2026 BMW X5 (Score: 88.0)

Price	Market Avg	Mileage
\$58,073	\$69,135	8,000 mi

Demo vs Live Mode

	Demo (default)	Live
Listings	Synthesized on-the-fly	Scraped via Bright Data
Scores	Simulated tiered discounts	Claude + SERP comparison
URLs	Live search links (real inventory)	Actual listing URLs
Tokens	Not required	Bright Data + Anthropic keys

Judges: `./dealscout/run.sh` → sidebar shows **pipeline v2**

Tech Stack

LangGraph

FastAPI

SSE

Streamlit

Bright Data MCP

Claude

SQLite

Python 3.12

Why this stack?

- LangGraph — explicit multi-agent orchestration with observable node events
- SSE — judges see agents work in real time, not a black box
- Bright Data MCP — production-grade scraping without building proxy infra
- No Docker — one-command setup for hackathon judges

Limitations & Future Work

- CarGurus entity IDs cover ~20 makes/models — others fall back to make-only search
- Demo listings are samples; links open filtered search, not exact VINs
- Live mode depends on site structure changes and rate limits

Next steps

- Expand entity ID map + auto-discovery from CarGurus price-trends URLs
- VIN-level matching and price history charts
- Email alerts when a scored deal appears in live inventory

Thank You

DealPulse Scout — find your best car deal in one query

```
./dealscout/run.sh  
curl localhost:8000/health  
http://localhost:8501
```

[Questions?](#)